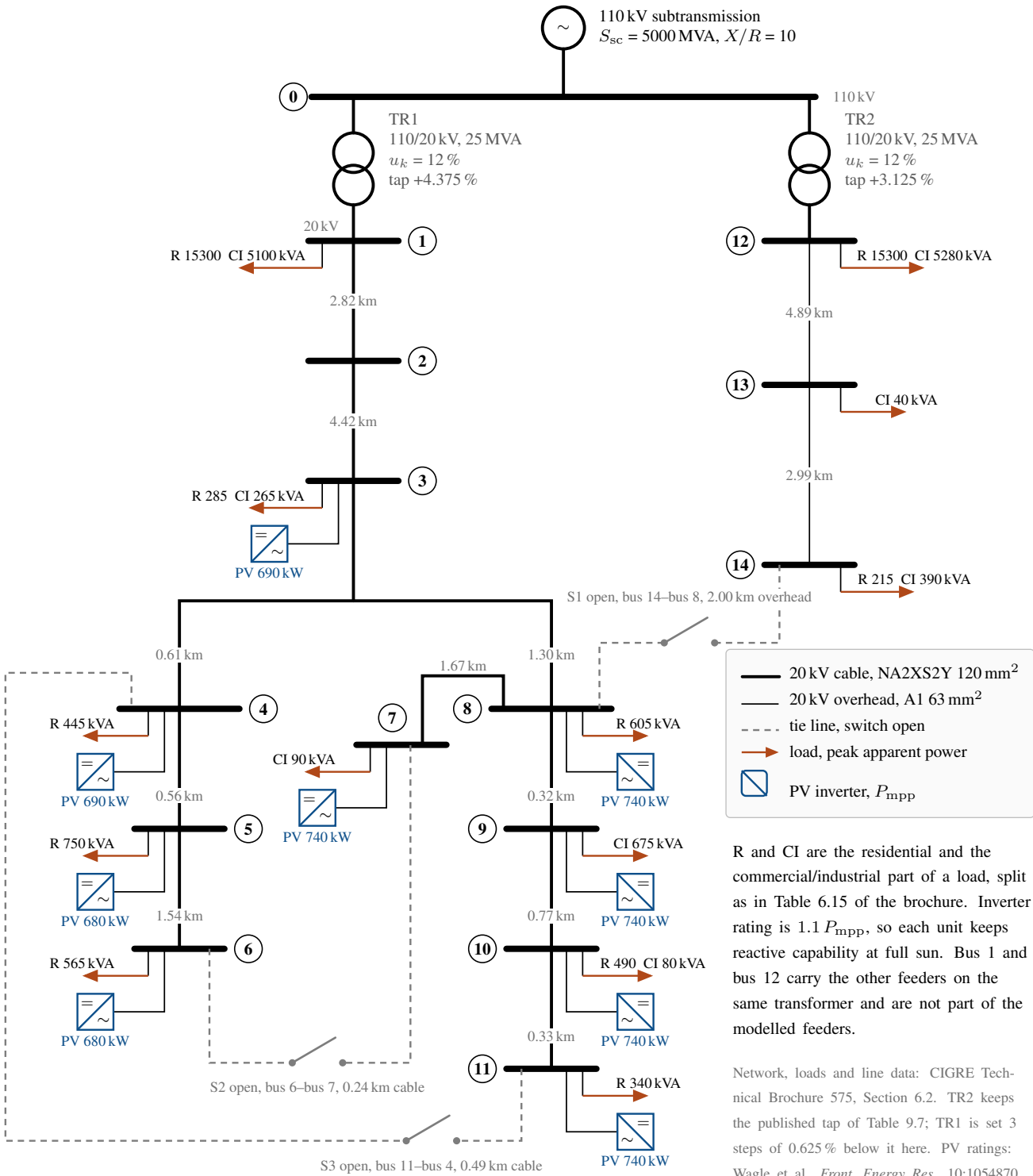


CIGRE European MV distribution benchmark

20 kV, 50 Hz, radial base case as studied here. Nine PV inverters on feeder 1, 6.44 MW in total.



R and CI are the residential and the commercial/industrial part of a load, split as in Table 6.15 of the brochure. Inverter rating is $1.1 P_{mpp}$, so each unit keeps reactive capability at full sun. Bus 1 and bus 12 carry the other feeders on the same transformer and are not part of the modelled feeders.

Network, loads and line data: CIGRE Technical Brochure 575, Section 6.2. TR2 keeps the published tap of Table 9.7; TR1 is set 3 steps of 0.625 % below it here. PV ratings: Wagle et al., *Front. Energy Res.* 10:1054870 (2023), Table 1. The 1.5 MW wind unit of the brochure at bus 7 is replaced by PV here.